



Cyclopia maculata and *Cyclopia subternata* (honeybush tea) inhibits adipogenesis in 3T3-L1 pre-adipocytes

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ARTICLE INFO

Keywords:

Cyclopia maculata
Cyclopia subternata
Polyphenols
3T3-L1 adipocytes
Obesity
Adipogenesis

ABSTRACT

The stems, leaves and flowers of *Cyclopia* have been consumed as a herbal tea 'honeybush tea' to treat various medical ailments since the 19th century. Plant polyphenols are reported to inhibit adipogenesis in cell and animal models of obesity. The aim of this study was to assess the effect of hot water extracts of two *Cyclopia* species, *C. maculata* and *C. subternata* on obesity in an *in vitro* model. The total polyphenol content of unfermented *C. subternata*, unfermented *C. maculata* and fermented *C. maculata* extracts was 25.6, 22.4 and 10.8 g GAE/100 g, respectively. The major compounds present in the extracts were: the flavonoid, phloretin-3',5'-di-C-glucoside in *C. subternata*, the xanthone, mangiferin in unfermented *C. maculata* and the flavanone, hesperidin in fermented *C. maculata*. All of the plant extracts inhibited intracellular triglyceride and fat accumulation, and decreased PPAR γ 2 expression. The higher concentrations of unfermented *C. maculata* (800 and 1600 μ g/ml) and *C. subternata* (1600 μ g/ml) were cytotoxic in terms of decreased mitochondrial dehydrogenase activity. Both fermented and unfermented *C. maculata*, at concentrations greater than 100 μ g/ml, decreased cellular ATP content. *Cyclopia maculata* and *C. subternata* inhibit adipogenesis *in vitro*, suggesting their potential as anti-obesity agents.

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Introduction

Obesity is increasing at alarming rates in both developed and developing countries, increasing the risk for insulin resistance, diabetes mellitus, coronary heart disease and hypertension (World Health Organisation, 2011). The disorder is characterised by enlargement of adipose tissue, caused by an increase in the number and size of adipocytes differentiated from fibroblastic pre-adipocyte precursors (Hsu and Yen, 2008). *In vitro* models, particularly murine 3T3-L1 pre-adipocytes which can be differentiated into mature adipocytes, have improved our understanding of the mechanisms involved in obesity. Inhibition of adipogenesis and restoration of adipocyte function are considered to be important anti-obesity mechanisms.

Polyphenols are gaining increased importance due to their beneficial effects on health. They are natural anti-oxidants that have the potential to treat a wide array of diseases, including obesity (Fraga et al., 2010; Meydani and Hasan, 2010). Their anti-obesity

effects have been demonstrated in *in vitro* models where they decrease adipogenesis and lipid accumulation and in rodent models of diet-induced obesity where decreased body weight, plasma triglycerides and hepatic steatosis was observed (Hsu and Yen, 2008). The demand for these anti-obesity preparations is increasing due to the adverse effects associated with current anti-obesity drugs (Li and Cheung, 2011) and the perception that natural products have less adverse effects and are safer to use than conventional therapeutics (Vermaak et al., 2011).

South African herbal teas, rooibos (*Aspalathus linearis*) and honeybush (*Cyclopia* spp.) are currently gaining popularity worldwide (Joubert et al., 2011; Joubert and De Beer, 2011) partly because scientific evidence of their beneficial health effects such as anti-oxidant, anti-cancer and anti-mutagenic properties (Joubert et al., 2008a, 2009) are mounting. Recent investigations of rooibos demonstrated that this herbal tea also has potential in alleviating obesity-related disorders (Beltran-Debon et al., 2011; Panti et al., 2011). In honeybush, the presence of high quantities of the xanthone glycoside mangiferin (Joubert et al., 2008b), a compound with anti-oxidant (Martinez et al., 2000), anti-inflammatory (Garrido et al., 2004), anti-diabetic (Ichiki et al., 1998) and anti-obesity effects (Guo et al., 2011; Yoshikawa et al., 2002) merits investigation of *Cyclopia* spp. The precursor of mangiferin, the

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benzophenone iriflophenone-3-C- β -glucoside that is present in *C. subternata* (Kokotkiewicz et al., 2012), inhibits triglyceride synthesis and the expression of associated genes in 3T3-L1 adipocytes (Zhang et al., 2011).

The genus *Cyclopia* is indigenous to the southeast and south-west coastal areas of South Africa. Records show that the stems, leaves and flowers of the plant have already been consumed as a herbal tea 'honeybush tea' in the 19th century (Du Toit et al., 1998; Joubert et al., 2011). More than 20 species of *Cyclopia* have been described with *Cyclopia genistoides*, *Cyclopia intermedia* and *Cyclopia subternata* enjoying commercial success (Joubert et al., 2011). To meet the global demand the commercial potential of other species such as *C. maculata* are considered. The aim of this study was to assess the effect of hot water extracts of two *Cyclopia* species, *C. maculata* and *C. subternata* on obesity in an *in vitro* model. Our investigation focused on a hot water extract of *C. subternata* and *C. maculata* as this type of extract of *Cyclopia* was previously shown to have anti-diabetic activity (Muller et al., 2011). 3T3-L1 pre-adipocytes were treated with the extracts during differentiation and the effect on adipogenesis and cell function was assessed by quantifying intracellular lipids, triglycerides and peroxisome proliferator-activated receptor (PPAR) gamma (γ) expression, while cell viability was assessed using the 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyl tetrazolium bromide (MTT) and adenosine triphosphate (ATP) assays.

Materials and methods

Materials

Mangiferin, gallic acids, glacial acetic acid, high performance liquid chromatography (HPLC) gradient grade acetonitrile, dexamethasone, 3-isobutyl-1-methylxanthine (IBMX), insulin, foetal bovine serum (FBS), tumour necrosis factor alpha (TNF α), bovine serum albumin (BSA), Dulbecco's modified Eagle's medium (DMEM) and phosphate buffered saline (PBS, pH 7.4) were obtained from Sigma–Aldrich (Steinheim, Germany). Formic acid, Folin-Ciocalteu reagent and 10% (v/v) formalin was purchased from Merck (Darmstadt, Germany). Phenolic compounds, eriocitrin, hesperidin, hesperetin and luteolin were purchased from Extrasynthese (Genay, France). Isomangiferin was isolated from *C. subternata* (De Beer et al., 2009). Nothofagin (phloretin-3'-C-glucoside) was obtained from the PROMEC unit of the Medical Research Council (MRC), South Africa. All other chemicals and reagents were analytical grade and purchased from Sigma–Aldrich or Merck. HPLC grade water was obtained by sequential purification through an Elix 5 UV (Millipore, Bedford, USA) and a Milli-Q academic water purifier (Millipore).

Plant material and extract preparation

Unfermented *C. maculata* extract was prepared from plant material (ca 25 kg) harvested from a natural population on a farm in the Riviersonderend area (Overberg) of the Western Cape Province of South Africa. The coarse, thick stems were removed before cutting into small pieces ("tea bag cut"), followed by drying at 40 °C to less than 10% moisture content (m/m) in a pilot-scale drying tunnel with cross-flow air movement. The dried plant material was coarse sieved to remove the fraction > 8 mesh. Laboratory-scale extraction of 150 g dried, sieved plant material with 1500 ml of freshly boiled purified water, entailed placing the container in a water bath at 93 °C and stirring the contents every few minutes to facilitate extraction over a 30 min period. The extract was filtered through a fine mesh cloth (Polymon, Swiss Silk Bolting Cloth

Mfg. Co., Zurich, Switzerland) and cooled to room temperature. Several batches were prepared and pooled before freeze-drying in a semi-pilot freeze-drier.

For preparation of fermented *C. maculata* extract and unfermented *C. subternata* extract industrial facilities were used as large quantities of the extracts were required for several studies. Approximately 1000 kg of *C. maculata* plant material was harvested from a natural population, growing on a farm in the Riversdale area (Overberg). This was fermented, dried and sieved (ca 300 kg; particle size < 4 mm) by a commercial tea processor. The plant material was extracted in a stirred vessel with 3000 l hot purified water, pre-heated to approximately 93 °C. The mixture was continuously stirred for 30 min, where after it was coarse filtered through screen and bag filters, cooled, micro-filtered and subjected to reverse osmosis. The semi-concentrated extract was further concentrated in a flow-through thin-film vacuum evaporator before being spray-dried. During spray-drying, the concentrate was exposed to an inlet temperature of 210 °C and an outlet temperature of 95 °C.

Unfermented *C. subternata* plant material (400 kg) was obtained from a commercial tea processor and batch extracted using a percolator type extraction vessel. Purified water preheated to 93 °C was introduced into the percolator from the top at a 1:10 (m/v) ratio and the resulting extract circulated for 35 min. The final extract was drained, centrifuged and concentrated under vacuum using a plate evaporator before vacuum-drying at 40 °C for 24 h.

Phenolic analysis

The total polyphenol content of the extract was determined using the method described by Singleton and Rossi (1965) with minor modifications in accordance with Arthur et al. (2012). Quantification of the major phenolic compounds in the extract was performed by high performance liquid chromatography with diode-array detection (HPLC-DAD) (De Beer and Joubert, 2010), using an Agilent 1200 system (Agilent technologies, Waldbron, Germany). Gradient separation was achieved with acetonitrile and 2% acetic acid at 1 ml/min flow rate on a Zorbax Eclipse XDB-C18 column (4.6 $\times$ 150 mm, 5 μ m particle size) from Agilent Technologies, temperature-controlled at 30 °C. The injection volumes were 10 and 20 μ l for analysis of the extract (6 mg/ml dissolved in purified water) and 10 μ l for the standard calibration mixtures. UV-vis spectra and retention times of the major peaks were compared to that of authentic standards for tentative identification. Mangiferin, hesperidin, hesperetin and eriocitrin were quantified using calibration curves for authentic reference standards, while isomangiferin, iriflophenone-3-C-glucoside, eriodictyol-glucoside, phloretin-3',5'-di-C-glucoside and scolymoside were quantified as mangiferin, hesperidin, eriocitrin, phloretin-3'-C-glucoside and luteolin equivalents, respectively (Arthur et al., 2012; Singleton and Rossi, 1965).

Analysis of the extract by HPLC-DAD coupled to mass spectrometric detection (MS) allowed further confirmation of peak identity. An Acquity UPLC system equipped with a binary solvent manager, sample manager and diode-array detector (Waters, Milford, MA, USA) was used in conjunction with a Synapt G2 Q-TOF system (Waters). Separation was performed as for HPLC-DAD, except that 2% acetic acid was replaced by 0.1% formic acid to reduce suppression of compound ionisation in the MS. The MS was operated with an electrospray ionisation source to obtain negative ionisation in high resolution mode (150–1000 amu) with MS^E (automated MS and MS/MS data acquisition). The MS was calibrated using sodium formate, while leucine enkephalin was used as lockspray. Operating parameters were as follows: capillary voltage, 2.5 kV; cone voltage, 15 V; source temp, 120 °C; desolvation temp, 275 °C, cone gas, N₂ at 50 l/h; desolvation gas N₂ at 650 l/h. The presence or absence of compounds previously identified in *Cyclopia*

spp. was confirmed based on literature data (Ferreira et al., 1998; Kamara et al., 2003, 2004; Kokotkiewicz et al., 2012)

Preparation of extracts for assays

The plant extracts or positive control (TNF α) used on days 1 to 7 were freshly reconstituted in DMEM containing 10% FCS, while on day 8 they were reconstituted in DMEM containing 0.1% (w/v) BSA without phenol red due to the interference of phenol red and serum in subsequent assays. Stock concentrations of 1.6 mg/ml were filter-sterilized using a low protein binding affinity 0.22 μ m syringe filter (Merck Millipore, Billerica, MA, USA) before dilution in growth medium. The medium containing the extract was warmed for at least 20 min at 37 °C prior to being added to the cells.

Cell culture and adipocyte differentiation

3T3-L1 mouse pre-adipocytes (CL-173, American Type Culture Collection, Manassas, VA, USA) were seeded in 96-well plates at a density of 2×10^4 cells/ml or in 6-well plates at a density of 2×10^5 cells/ml. Cells were grown to confluence in DMEM containing 10% FCS at 37 °C, 5% CO $_2$ in humidified air. Twenty four hours post visual confluence, cell differentiation was induced by culturing in adipogenesis inducing medium (DMEM containing 10% FCS, 0.6 μ M dexamethasone, 0.1 μ M IBMX and 16 μ M insulin) for five days with daily refreshing. Cells were then cultured in adipogenesis maintenance medium (DMEM containing 10% FCS and 16 μ M insulin) for three days with daily refreshing. The effect on adipogenesis was assessed by adding various concentrations of the plant extracts (0 to 1600 μ g/ml) to the culture media during differentiation. TNF α , a known inhibitor of adipogenesis (Cawthorn et al., 2007), was used as a positive control in all assays.

Intracellular lipid accumulation

3T3-L1 pre-adipocytes were differentiated in 6-well plates and treated with the extracts as described previously. After differentiation, the cells were washed with PBS and fixed with 10% (v/v) formalin for 1 h. Thereafter cells were stained with 1% (v/v) Oil Red O in isopropanol for 30 min and washed with deionized water thrice. The stained oil droplets were solubilised by incubating with isopropanol for 15 min and absorbance, an indication of lipid accumulation, was determined at 570 nm on a BioTek® ELX 800 plate reader (BioTek Instruments Inc., Winooski, VT, USA).

Triglyceride content

After differentiation and treatment with extracts in 96-well plates, 3T3-L1 adipocytes were washed with PBS and intracellular triglycerides were quantified using a triglyceride kit according to the manufacturer's instructions (Biovision Inc., Milpitas, CA, USA). The kit digests triglycerides into free fatty acids and glycerol, the latter of which was measured using a BioTek® FLX 800 plate reader (Mosmann, 1983).

Cytotoxicity

A method adapted from Mosmann (1983) was used to determine mitochondrial dehydrogenase activity. Briefly, cells treated in 96-well plates were incubated with 2 μ g/ml of MTT in PBS for 1 h at 37 °C, and the produced formazan dissolved in 200 μ l DMSO and 25 μ l Sorensen's phosphate buffer (pH 10.5) before quantified at 570 nm. The amount of ATP generated by cells was quantified using the CellTiter-Glo® Luminescent Cell Viability Assay kit (Promega Corporation, Madison, WI, USA) according to the manufacturer's

guidelines. Luminescence, indicative of ATP content, was quantified on a BioTek® FLX 800 plate reader.

Western blot analysis

To investigate the effect of treatment on PPAR γ expression, experiments were scaled up to 6-well plates, and pre-adipocytes were differentiated in the presence of 80 μ g/ml of each of the three extracts. This concentration was selected because it inhibited adipogenesis without being cytotoxic (no reduction in mitochondrial dehydrogenase activity or ATP content). 3T3-L1 adipocytes were rinsed with pre-warmed PBS, whereafter lysis buffer (50 mM Tris pH 7.5, 1 mM DTT, 50 mM NaF, 100 μ M Na $_3$ VO $_4$, 1% (v/v) NP-40 (NP40), 1% (v/v) Triton X 114, 25 μ g/ml RNase, 1X protease and phosphatase inhibitor tablets (Roche Diagnostics, Basel, Switzerland) and 1 mM phenylmethylsulfonyl fluoride (PMSF)) was added and cells were dislodged using a cell scraper. Thereafter cells were homogenised in a TissueLyser (Qiagen, Hilden, Germany) at 25 Hz for 1 min, followed by cooling on ice for 1 min. This step was repeated thrice. Cell lysates were clarified by centrifugation at 12,000 g for 10 min at 4 °C, and protein concentration determined using the RC DC kit (Biorad Laboratories Inc., Hercules, USA). Forty micrograms of protein was separated by electrophoresis on 10% polyacrylamide gels containing sodium dodecyl sulphate (SDS) and transferred to polyvinylidene fluoride (PVDF) membranes (Pierce, Rockford, USA). Membranes were blocked with 5% (w/v) non-fat milk for 6 h, and then incubated with PPAR γ or β -tubulin (Cell Signalling, Danvers, MA, USA) diluted 1:1000 at 4 °C overnight. Membranes were incubated with a horseradish peroxidase (HRP)-conjugated secondary antibody for 1 h at room temperature. Immunodetection was conducted using the LumiGLO Reserve kit (Santa Cruz, Santa Cruz, CA, USA) according to the manufacturer's instructions. Immunoreactive proteins were visualised and quantified using a ChemiDoc image analyser (BioRad Laboratories).

Enzyme-linked immunosorbant assay

Adiponectin and leptin secretion from adipocytes cultured in 6-well plates was assessed in culture supernatants using enzyme-linked immunosorbant assay (ELISA) kits (Merck Millipore), according to the manufacturer's instructions. Protease and phosphatase inhibitor cocktail tablets (Roche Diagnostics) were added to the culture supernatants according to the manufacturer's recommendations.

Statistical analysis

All experiments were done as three replicates in three independent experiments. Data are expressed as the mean $\pm$ standard deviation. Statistical significance between groups was determined by one-way analysis of variance (ANOVA) and the Dunnett *post hoc* test (Graphpad Prism® version 5.02). A *p* value of less than 0.05 (*p* < 0.05) was considered to be statistically significant.

Results

Phenolic content

Unfermented *C. subternata*, unfermented *C. maculata* and fermented *C. maculata* contained 25.6, 22.4 and 10.8 g (gallic acid equivalents (GAE)/100 g) polyphenols, respectively (Table 1). In addition to having a higher polyphenolic content, unfermented extracts also contained a wider variety of phenolic compounds. HPLC-DAD and UPLC-DAD-MS analysis showed that the major compounds present were: the flavonoid, phloretin-3',5'-di-C-glucoside

Table 1
Polyphenolic composition (g/100 g) of the extracts.

Parameter	Fermented <i>C. maculata</i>	Unfermented <i>C. maculata</i>	Unfermented <i>C. subternata</i>
Total polyphenols ^a	10.81	22.43	25.59
Iriflophenone-3-C-glucoside ^b	nd	1.13	0.37
Mangiferin	0.54	6.19	0.59
Isomangiferin	0.55	2.08	0.51
Eriodictyol-glucoside ^c	nd	nd	0.52
Eriocitrin	nd	0.42	0.50
Scolymoside ^d	Traces ^e	Traces ^e	0.46
Phloretin-3',5'-di-C-glucoside ^f	nd	0.17	1.10
Hesperidin	0.66	0.80	0.65
Hesperetin	0.06	nd	nd

^a Gallic acid equivalents.

^b hesperidin equivalents.

^c eriocitrin equivalents.

^d luteolin equivalents.

^e not quantifiable due to low peak area and co-elution with unidentified compound.

^f phloretin-3'-C-glucoside equivalents.

in *C. subternata*, the xanthones, mangiferin and isomangiferin in unfermented *C. maculata* and the flavanone rutinoside, hesperidin in fermented *C. maculata*. Hesperidin was also present in high amounts in unfermented *C. maculata* and *C. subternata*. A small quantity of its aglycone, hesperetin, was also present in fermented *C. maculata*. Combined the xanthones comprised more than 8% of unfermented *C. maculata*, and 1% of each fermented *C. maculata* and *C. subternata*. The isomangiferin content of the extracts are, however, overestimated due to co-elution with an unidentified compound. The HPLC-DAD chromatograms (Fig. 1) and HPLC-DAD-MS chromatograms (data not shown) also shows the presence of several unidentified peaks.

C. maculata and *C. subternata* inhibits adipocyte differentiation, triglyceride accumulation and PPAR γ 2 expression

Oil Red O staining showed that differentiation of 3T3-L1 pre-adipocytes with the plant extracts reduced intracellular lipid accumulation. The different extracts exhibited similar effects on lipid accumulation (Fig. 2a). The higher concentration of the positive control (10 ng/ml), TNF α , reduced intracellular lipid accumulation by 36% ($p < 0.001$).

Only the higher concentrations of the plant extracts ($>100 \mu\text{g/ml}$) decreased intracellular triglyceride accumulation compared to the untreated control (Fig. 2b). Fermented *C. maculata* reduced triglyceride accumulation by 23% at 200 $\mu\text{g/ml}$ ($p < 0.05$), 42% at 400 $\mu\text{g/ml}$ ($p < 0.001$), 52% at 800 $\mu\text{g/ml}$ ($p < 0.001$) and 27% at 1600 $\mu\text{g/ml}$ ($p < 0.01$). Treatment with unfermented *C. maculata* caused a 29% reduction of triglycerides at 100 $\mu\text{g/ml}$ ($p < 0.05$), 36% at 200 $\mu\text{g/ml}$ ($p < 0.01$), 48% at 400 $\mu\text{g/ml}$ ($p < 0.001$), 55% at 800 $\mu\text{g/ml}$ ($p < 0.001$), 37% at 1600 $\mu\text{g/ml}$ ($p < 0.01$). *C. subternata* caused a 38% inhibition of lipid accumulation at 400 $\mu\text{g/ml}$ ($p < 0.01$), 41% at 800 $\mu\text{g/ml}$ ($p < 0.001$) and 57% at 1600 $\mu\text{g/ml}$ ($p < 0.001$). TNF α did not effect intracellular triglyceride content.

Western blot analysis was used to quantify the expression of PPAR γ in 3T3-L1 adipocytes differentiated with the plant extracts. PPAR γ exists as two isoforms; with isoform 2 being the predominant form expressed in adipocytes (Auwerx, 1999). Treatment with the three extracts and the positive control, TNF α , decreased the expression of PPAR γ isoform 2, while only *C. subternata* increased the expression of isoform 1 (Fig. 3). The expression of PPAR γ isoform 2 was decreased by 43% ($p < 0.01$), 59% ($p < 0.001$), 61% ($p < 0.001$) and 30% ($p < 0.05$) by fermented and unfermented *C. maculata*, *C. subternata* and TNF α , respectively. The expression of

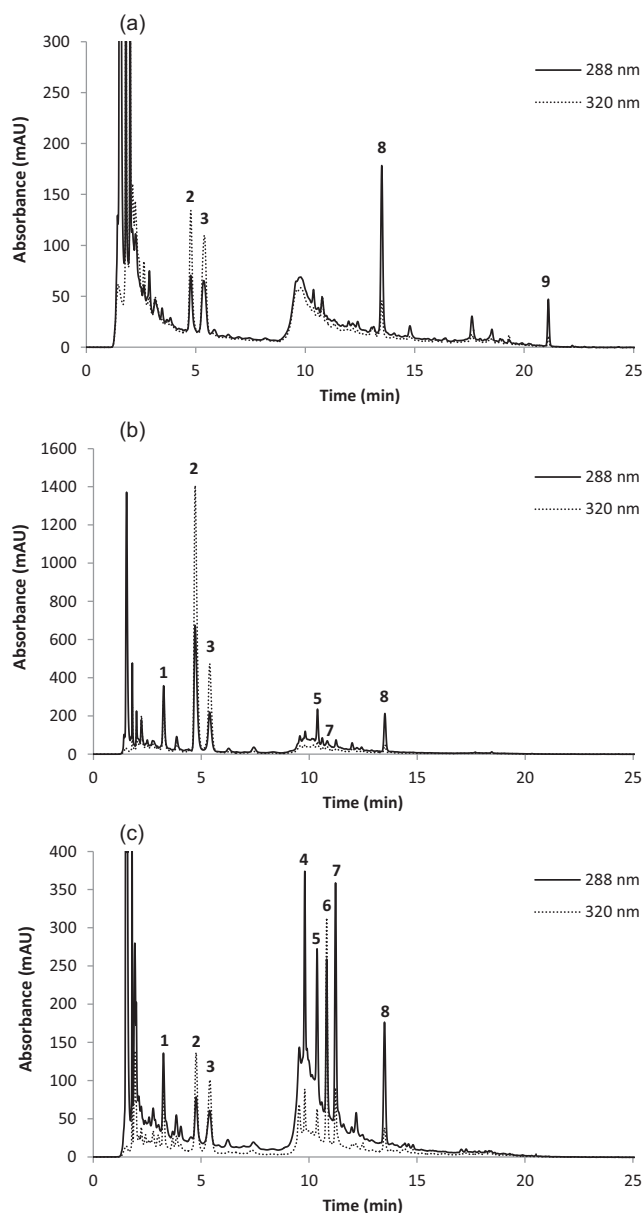


Fig. 1. HPLC chromatogram of (a) fermented *C. maculata*, (b) unfermented *C. maculata* and (c) unfermented *C. subternata* extracts [1, iriflophenone-3-C-glucoside; 2, mangiferin; 3, isomangiferin; 4, eriodictyol-glucoside; 5, eriocitrin; 6, scolymoside; 7, phloretin-3',5'-di-C-glucoside; 8, hesperidin; 9, hesperetin].

isoform 1 was upregulated by 99% ($p < 0.01$) after treatment with *C. subternata*.

Treatment increases adiponectin and leptin secretion

Differentiation of 3T3-L1 pre-adipocytes with fermented *C. maculata* increased adiponectin secretion >900 -fold ($p < 0.001$) (Table 2). Neither TNF α nor any of the other extracts increased adiponectin secretion. Leptin secretion was increased 1.9-fold ($p < 0.001$), 2-fold ($p < 0.001$) and 1.7-fold ($p < 0.01$) by fermented and unfermented *C. maculata*, and *C. subternata*, respectively (Table 2). Differentiation with TNF α did not affect leptin secretion.

Effect of plant extracts on cytotoxicity

To investigate whether *C. maculata* or *C. subternata* exhibits cytotoxicity, mitochondrial dehydrogenase and metabolic activity

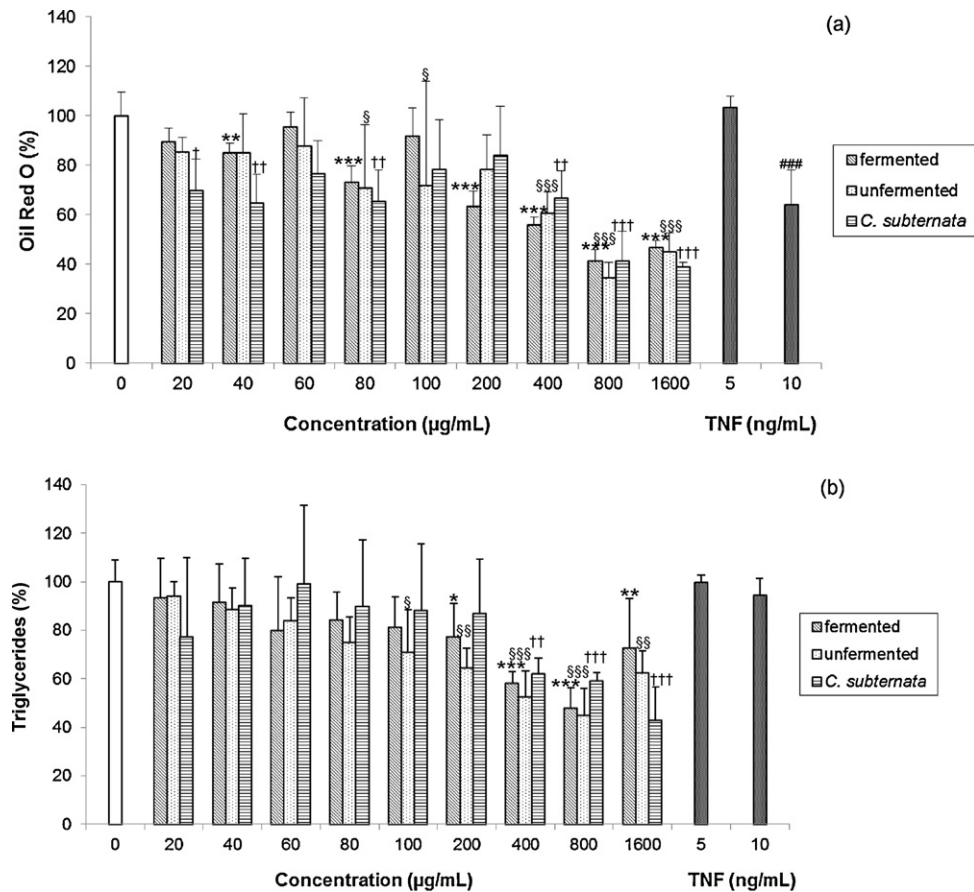


Fig. 2. Effect of plant extracts on lipid accumulation. 3T3-L1 pre-adipocytes were differentiated with fermented or unfermented *C. maculata*, unfermented *C. subternata* or TNFα for 8 days. Intracellular lipid accumulation was quantified by Oil Red O staining (a) or with a commercial triglyceride quantification kit (b) and expressed as a percentage of lipid accumulation in untreated cells. Data are expressed as the mean ± standard deviation of three independent experiments, each performed in triplicate. *§,† $p < 0.05$ vs untreated control. **§§,†† $p < 0.01$ vs untreated control. ***§§§,†††,†††† $p < 0.001$ vs untreated control.

was assessed using the MTT and ATP assays. Recent concerns that polyphenolic compounds could interfere with the MTT assay (Wang et al., 2010; Wisman et al., 2008), prompted us to evaluate cytotoxicity using the ATP assay also.

None of the concentrations of fermented *C. maculata* used during the study decreased mitochondrial dehydrogenase activity (Fig. 4). Unfermented *C. maculata* decreased mitochondrial dehydrogenase activity by 39% at 800 µg/ml ($p < 0.05$) and by 57% at

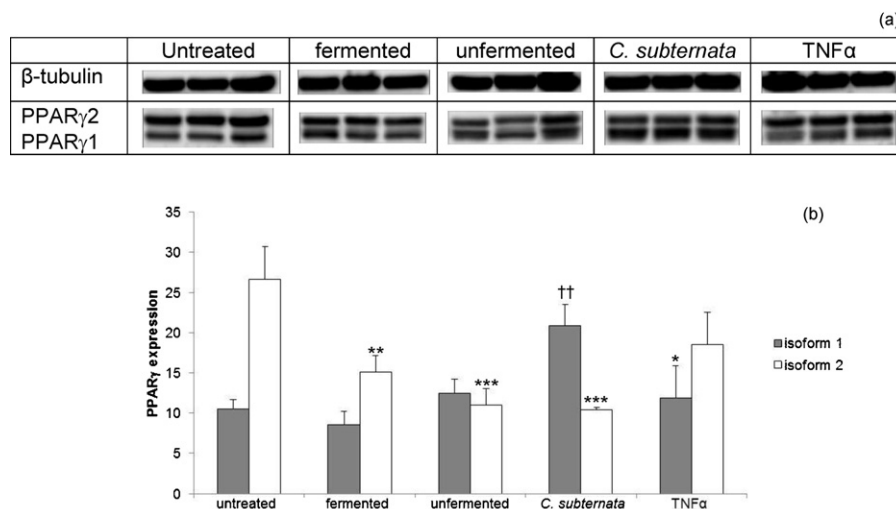


Fig. 3. Effect of plant extracts on PPARγ expression. 3T3-L1 adipocytes were differentiated in the presence of 80 µg/mL of fermented or unfermented *C. maculata*, unfermented *C. subternata* or 10 ng/mL of TNFα for 8 days, whereafter protein lysates were analysed by Western blotting. (a)–Densitometry images of the protein bands detected. (b)–Protein expression normalised to β-tubulin. Data is presented as the mean ± standard deviation of three independent experiments, each performed in triplicate. * $p < 0.05$ vs untreated control. ** $p < 0.01$ vs untreated control. *** $p < 0.001$ vs untreated control.

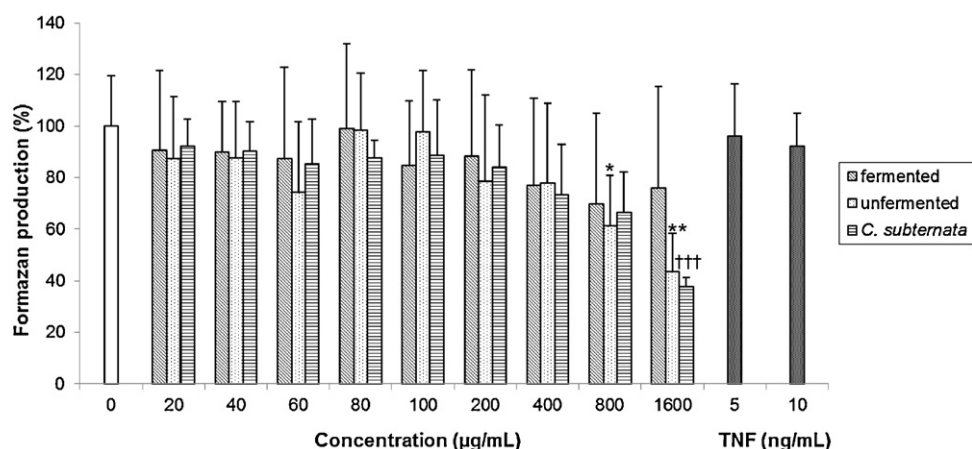


Fig. 4. Effect of plant extracts on mitochondrial dehydrogenase activity. 3T3-L1 pre-adipocytes were differentiated with fermented or unfermented *C. maculata*, unfermented *C. subternata* or TNF α for 8 days. MTT values were calculated as a percentage of untreated cells. Data is expressed as the mean $\pm$ standard deviation of three independent experiments, each performed in triplicate. * $p < 0.05$ vs untreated control. ** $p < 0.01$ vs untreated control. *** $p < 0.001$ vs untreated control.

1600 $\mu\text{g/ml}$ ($p < 0.01$). *C. subternata* reduced mitochondrial dehydrogenase activity by 62% at 1600 $\mu\text{g/ml}$ ($p < 0.001$) only. TNF α did not affect mitochondrial dehydrogenase activity.

ATP content was decreased by 35% ($p < 0.05$) and 34% ($p < 0.05$) at 100 $\mu\text{g/ml}$, 37% ($p < 0.05$) and 47% ($p < 0.001$) at 200 $\mu\text{g/ml}$, 51% ($p < 0.01$) and 62% ($p < 0.001$) at 400 $\mu\text{g/ml}$ and 68% ($p < 0.01$) and 74% ($p < 0.001$) at 800 $\mu\text{g/ml}$ by fermented and unfermented *C. maculata*, respectively (data not shown). Treatment with *Cyclopia subternata* and TNF α did not affect ATP content.

Discussion

Obesity is a growing epidemic worldwide and significantly increases the risk of developing a number of chronic diseases such as insulin resistance, diabetes mellitus, coronary heart disease, hypertension (World Health Organisation, 2011). The currently available anti-obesity drugs are plagued by numerous adverse effects (Li and Cheung, 2011), renewing interest in natural products as therapeutics since they are considered safer than their synthetic counterparts (Vermaak et al., 2011). Herbal teas such as *Cyclopia* spp. are gaining increased importance due to their beneficial effects on health (Joubert et al., 2008a, 2009). The ability to inhibit adipogenesis, *in vitro* or *in vivo*, is used to evaluate phytochemicals for anti-obesity potential (Hsu and Yen, 2008). In this study, 3T3-L1 pre-adipocytes were differentiated in the presence of hot water extracts of *C. maculata* and *C. subternata* to assess their anti-obesity potential.

All three extracts, at most of the concentrations tested, decreased lipid and triglyceride accumulation. At the molecular

level, expression of PPAR γ isoform 2 was decreased after treatment. PPAR γ is a nuclear receptor and transcription factor regulating lipid and glucose metabolism and is often referred to as the 'master regulator' of adipogenesis (Auwerx, 1999). The protein exists as two isoforms, PPAR γ 1 and PPAR γ 2, where isoform 1 is the predominant form and is ubiquitously expressed, while isoform 2 is more abundant in adipose tissue and is considered the key regulator of adipogenesis. Previously, Vidal-Puig et al. (1996) showed that high fat feeding of obese mice increased PPAR γ 2 mRNA levels, but had no effect on PPAR γ 1 expression. Differences in expression levels of PPAR γ 1 and 2 proteins were observed in this study. Future studies to investigate the biological differences between these two isoforms would be of interest. Furthermore, a number of studies have linked downregulation of lipogenesis by polyphenols to increased AMP-activated protein kinase (AMPK) activity (Hwang et al., 2009; Niu et al., 2012; Zhang et al., 2011). AMPK controls many metabolic processes including glucose, lipid and cholesterol metabolism, thus future studies investigating the effect of these extracts on AMPK are warranted.

Fermentation decreased polyphenolic content more than 50%. The effects of fermentation (the oxidation of phenolic compounds required for the development of the characteristic sweet aroma and flavour of honeybush tea) on lowering polyphenolic content, especially mangiferin, and decreasing anti-oxidant activity has been previously reported for *Aspalathus linearis* and other *Cyclopia* spp. including *C. subternata* (Joubert et al., 2008b). This study was the first investigation of *C. maculata*, thus both the fermented and unfermented extract was analysed. Both unfermented and fermented *C. maculata* displayed anti-adipogenic properties, suggesting that fermentation does not affect the ability to inhibit fat and triglyceride accumulation. These results are promising since fermented plant extracts are preferred due to their sensory properties. Moreover, the fermented extract was less cytotoxic, in terms of mitochondrial dehydrogenase activity and ATP content, than the unfermented 'green' extract.

Combined the xanthenes, mangiferin and isomangiferin, comprised more than 8% of unfermented *C. maculata*, and 1% of each fermented *C. maculata* and *C. subternata*. Mangiferin was first isolated from *Mangifera indica* (mango tree) by Wiechowski in 1908. The roots, bark, leaves, flowers and fruit of the tree have traditionally been used to treat various ailments. Studies have demonstrated a number of health effects, including anti-microbial (Singh et al., 2009), anti-oxidant (Martinez et al., 2000), anti-inflammatory (Garrido et al., 2004), anti-diabetic (Giron et al., 2009; Ichiki et al., 1998) and anti-obesity effects (Guo et al., 2011; Niu

Table 2
Effect of plant extracts on adiponectin and leptin secretion.^a

Treatment	Adiponectin	Leptin
Untreated	100% $\pm$ 43%	100% $\pm$ 0%
Fermented <i>C. maculata</i>	93,053% $\pm$ 2346%***	193% $\pm$ 12%***
Unfermented <i>C. maculata</i>	88% $\pm$ 78%	197% $\pm$ 0%***
Unfermented <i>C. subternata</i>	42% $\pm$ 34%	173% $\pm$ 0%**
TNF α	193% $\pm$ 26%	120% $\pm$ 1%

^a 3T3-L1 pre-adipocytes were differentiated in the presence of 80 $\mu\text{g/ml}$ extract or 10 ng/ml of TNF α for 8 days. The secretion of adiponectin and leptin in cell culture supernatants was quantified by ELISA using commercial kits. Values were calculated as a percentage of adiponectin or leptin secretion of untreated cells. Data is expressed as the mean $\pm$ standard deviation of three independent experiments, each performed in triplicate.

** $p < 0.01$ vs untreated control.

*** $p < 0.001$ vs untreated control.

et al., 2012; Yoshikawa et al., 2002). Mangiferin and its precursor, the benzophenone iriflophenone-3-C- β -glucoside that is present in *C. subternata* (Kokotkiewicz et al., 2012) inhibited triglyceride synthesis and the expression of associated genes in 3T3-L1 adipocytes (Zhang et al., 2011). Mangiferin is commercially available as an anti-oxidant and anti-diabetic supplement under the brand names Vimang® (Cuba) and Salaretin® (Sri Lanka), respectively.

The flavanone glycoside, hesperidin, was present in high amounts in unfermented and fermented *C. maculata* and *C. subternata*. Hesperidin is most common in citrus fruits and has a number of health benefits including anti-oxidant and anti-inflammatory (Emim et al., 1994), and anti-obesity (Bok et al., 1999; Chiba et al., 2003; Park et al., 2001) activity. Chiba et al. (2003) demonstrated that hesperidin treatment is able to inhibit bone loss, and decrease hepatic and serum lipids in ovariectomized mice (Chiba et al., 2003).

Phloretin-3',5'-di-C-glucoside was the most abundant phenolic compound present in *C. subternata*, constituting more than 1% of the extract. The flavonoid was present in low amounts or undetected in *C. maculata*. Phloridzin (Phloretin-2'-O-glucoside) and phloretin is found in high quantities in apples (Escarpa and Gonzalez, 1998), and a number of studies have demonstrated anti-diabetic (Dimitrakoudis et al., 1992; Najafian et al., 2012; Rastogi et al., 1997; Rossetti et al., 1987) and anti-obesity (Najafian et al., 2012) effects. Treatment of 3T3-L1 pre-adipocytes with both *C. maculata* and *C. subternata* decreased the expression of PPAR γ 2, while only *C. subternata* increased the expression of PPAR γ 1. Differences in ATP content were also observed after treatment with the different extracts. *C. maculata* at concentrations $\geq 100 \mu\text{g/ml}$ decreased ATP content, whereas *C. subternata* had no effect. PPAR γ regulates lipid and glucose metabolism, and decreased expression of its isoforms PPAR γ 1 and PPAR γ 2 is observed during insulin deficient diabetes (Vidal-Puig et al., 1996). Thiazolidinediones and mangiferin improve insulin resistance and stimulates glucose uptake by activating PPAR γ 1 and PPAR γ 2 (Giron et al., 2009; Vidal-Puig et al., 1996). The increased expression of PPAR γ 1 after treatment with *C. subternata* is not due to mangiferin alone since unfermented *C. maculata* contains higher amounts of mangiferin than *C. subternata*. However, the role of other compounds present in the extract, due to possible synergistic effects must be considered. As recently reviewed (Ulrich-Merzenich et al., 2009; Wagner and Ulrich-Merzenich, 2009), synergism between bioactive components in plant extracts could result in more effective treatment of obesity, a multifactorial disorder that involves the interaction of complex cellular networks. Fractionation of these extracts and evaluating their anti-obesity and anti-diabetic effects (insulin resistance and glucose uptake) will allow more insight about the roles of PPAR γ 1 and PPAR γ 2.

It is essential that potential anti-obesity agents inhibit adipogenesis without being cytotoxic. Cell viability was measured using the MTT and ATP assays, two methods that are commonly used to assess *in vitro* toxicity. Both methods measure metabolic activity of cells; the MTT assay measures mitochondrial dehydrogenase activity, while the ATP assay measures cellular ATP (Ulukaya et al., 2008; Wang et al., 2010). Both assays showed that the plant extracts were not cytotoxic at concentrations ranging between 0 and $100 \mu\text{g/ml}$. Differences were noted when using *C. maculata* (fermented and unfermented) concentrations between 100 and $800 \mu\text{g/ml}$; the MTT assay showed non-cytotoxicity, while a decrease in ATP was observed, suggesting cytotoxicity. Similar differences between the two assays were reported previously for other polyphenol-rich extracts (Wang et al., 2010; Wisman et al., 2008). These results emphasize the importance of selecting the correct test to evaluate cytotoxicity.

Treatment with fermented *C. maculata* increased adiponectin secretion, while treatment with all three extracts increased

leptin secretion. Adiponectin and leptin are adipokines with health promoting effects. Adiponectin has anti-atherogenic, anti-inflammatory and anti-diabetic (Pajvani and Scherer, 2003). Decreased expression of adiponectin is associated with obesity and insulin resistance. Leptin, previously labelled the 'satiety or anti-obesity hormone', inhibits food intake and stimulates thermogenesis (Ahima and Flier, 2000). A number of other plants with anti-obesity potential mediated by decreased PPAR γ expression, as observed in this study, also regulate adiponectin and leptin secretion (Oben et al., 2008; Seo et al., 2011).

Both *C. maculata* and *C. subternata* inhibited adipogenesis without being cytotoxic, suggesting their potential as anti-obesity agents. This is the first study to illustrate that *Cyclopia* has anti-obesity properties. Our *in vitro* findings are currently being validated in a Wistar rat model of diet-induced obesity.

Conflict of Interest

The authors have no conflict of interest to disclose.

Acknowledgements

The authors would like to thank the Cape Honeybush Tea Company, Mossel Bay, South Africa for preparation of the fermented *C. maculata* and unfermented *C. subternata* extracts. The research was supported by the Indigenous Knowledge Systems Programme of the National Research Foundation, funded by the Department of Science and Technology, South Africa.

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